

Guided Self-Study: Parkinson's Disease

Introduction

This guided self-study is designed to teach you about the symptoms, etiology, and medical treatments associated with **Parkinson's Disease (PD)**. PD is a progressive neurodegenerative disease that is estimated to affect nearly 6.1 million people worldwide. Most individuals with PD will be diagnosed in their 60s. However, there are some instances where PD can develop in the late 20s. This is termed “Young-onset” PD and it is what actor and director Michael J Fox was diagnosed with in 1991 at the age of 29 – the peak of his acting career.

In this activity you are taking the role of a medical intern or advanced trainee. Imagine that you are part of a clinical neuroscience team working under a neurology attending. In this role, you are helping to gather information, ask questions, and review diagnostic data with guidance.

During Part I you will be interacting with an AI character “James Patient”. You will be asking “James” a series of questions to assess their overall well-being and the specific symptoms they have been experiencing. During Part II we will discuss the brain areas involved and cover some of the cellular and molecular changes that occur in the brain. The activity for Part II is designed as an application-type problem. Here we will consider some modeled clinical data and make comparisons to healthy patients. Part III of the activity is designed to tie all of these ideas together and help you make sense of your discussion with James in light of the neuroimaging data.

Learning Objectives

1. **Describe** the motor and non-motor symptoms characteristic of Parkinson's Disease and **relate** these symptoms to changes in basal ganglia circuits.
2. **Interpret** clinical data to **evaluate** the integrity of dopamine systems in the brain and their role in the progression of Parkinson's disease.
3. **Explain** the mechanism of Levodopa and **evaluate** its therapeutic limitations based on clinical and neuroimaging evidence.

Part I. The Intake Interview

For this portion of the activity, you will be interacting with “James Patient” an AI Chatbot. I’ve designed this bot to model some of the characteristics of patients with early Parkinson’s disease. To access the bot follow this link ([click here](#)). You may need to make a free account in order to interact with the bot.

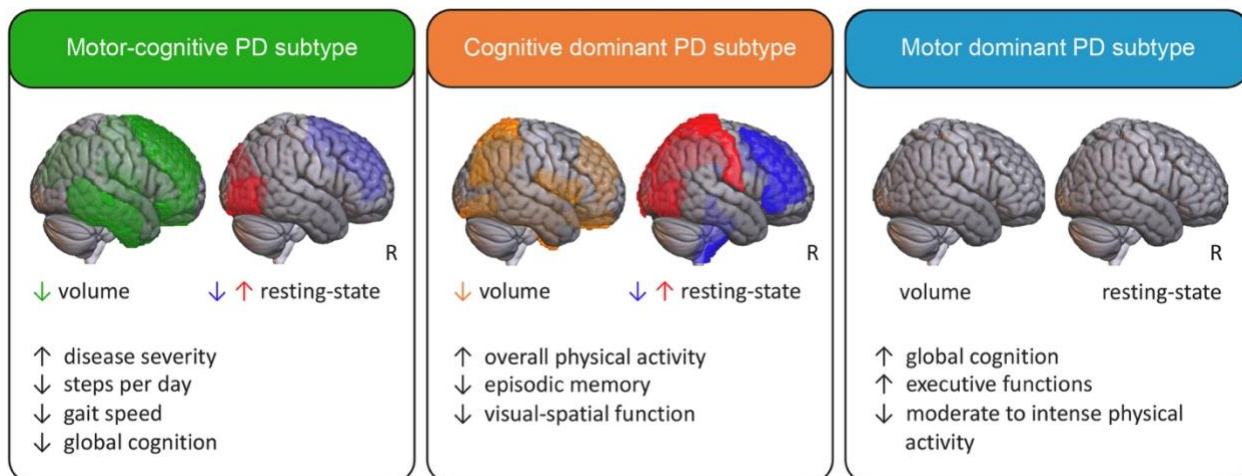
Since “James” is an AI bot, everyone will have a slightly different experience with James. However, the basics of what he has been feeling and experiencing will be the same. Below you have a set of basic questions. You can use these as guidelines to help structure your conversation with “James”. “James” may share more information, making some questions repetitive. It’s ok if you go off script, especially if something interesting comes up. Think like a clinician: gather information and build trust with your patient. Take notes along the way. There are 5 questions on the answer sheet that go along with this part of the exercise.

1. **Start the conversation.**
 - a. Hi James, can you tell me what brought you in today?
 - b. When did you first start noticing these changes?
2. **Ask about movements.**
 - a. Have you noticed any shaking, stiffness, or slowness?
 - b. Has it been harder to walk, move around, or keep your balance?
 - c. How has your handwriting or speech changed, if at all?
3. **Explore his daily life.**
 - a. What everyday tasks have become harder for you?
 - b. Is this affecting things like work, hobbies, or relationships?
4. **Ask about treatments.**
 - a. Have you tried any medications so far?
 - b. What did you notice after starting the medication?
 - c. Did it help? Any side effects?
5. **Check for non-motor symptoms.**
 - a. How have you been feeling emotionally – any changes in sleep or mood?
 - b. Has anything changed with digestion or appetite?
6. **Ask about family and psychological history.**
 - a. Do you have a family history of neurological conditions?
 - b. How has your support system been – family, friends, caregivers?
7. **Wrap-up**
 - a. Do you have any family history of similar issues?
 - b. What’s your biggest concern about all of this?

After you are done having your conversation with “James”, make sure to answer the questions on the answer sheet. Here you will summarize your findings.

Part II. Neuronal Basis of Motor Disturbances in PD

“James” is clearly having a hard time and a cluster of symptoms that cover motor and non-motor functions. This is pretty typical for a patient with PD. However, not all patients present with the same symptoms. PD, like many diseases is heterogeneous. Meaning, that symptoms can vary. This is especially true for early-stage PD where symptoms are starting to emerge. We can generally cluster PD symptoms into one of three categories based on their symptoms. This is shown in the image below (taken from Albrecht et al., Parkinsonism & Related Disorders (2022)).



As you can see, these disease groupings depend on the kinds of symptoms that the patient is suffering.

1. Motor-cognitive PD subtype – this subtype is categorized by symptoms that target their motor capabilities but also affect their cognitive function. From the figure above, we see that they have trouble moving and walking and their global cognition is also impacted. Global cognition is an all-in-one measure that describes executive functions like attention, orientation, memory, language, and decision making.
2. Cognitive dominant PD subtype – this subtype is categorized by cognitive impairments. These patients have noticeable cognitive decline where their memory is starting to become spotty and they begin struggling with visual-spatial information (like mentally orienting themselves or objects based on visual information, or judging distances based on visual perception). Interestingly, these

patients show hyperlocomotion. Their movements and physical activity is overall increased from normal. This is not something we usually associate with PD.

3. Motor dominant PD subtype – this subtype is associated with moderate to severe impairments in motor functions, but their cognitive and executive functions are fine or in some cases are even increased above normal.

Since PD is a progressive disease, we often see patients start at the Motor dominant PD subtype and eventually transition to a motor-cognitive PD subtype. Like Alzheimer's, PD generally gets worse and is not isolated to one brain area. But rather, grows to affect many different brain areas. This is what causes the cluster of motor and non-motor symptoms associated with PD. We could have a whole unit on all of the brain areas affected. However, we will focus on one today – the substantia nigra (SN).

PD is thought to start in a brain area known as the **substantia nigra (SN)**. This brain area is a part of a larger circuit known as the basal ganglia. The basal ganglia contributes to movement by modifying motor commands. It is the basal ganglia that helps to select for “appropriate” or “wanted” movement and turns off “unwanted” or “uncontrolled” movements. You can almost think of the basal ganglia as a bouncer. When a motor command is sent, it is the basal ganglia that serves as the first check, allows for the initiation of movement, and makes sure that only the chosen movement is initiated in the primary motor cortex.

Let's consider an example. You are walking on campus and spot a friend approaching from a distance. Without much conscious effort, you lift your arm and give them a smooth, friendly wave. While this seems simple, your brain just completed a complex set of negotiations between the premotor cortex, the basal ganglia, and the primary motor cortex. Let's see what happened in each region:

1. Premotor cortex identified what movement to perform (“Raise arm and wave”) and sent this information to the primary motor cortex and the basal ganglia.
2. The basal ganglia working as a gatekeeper evaluated the wave against other competing motor plans (“Don't flail. Don't swat at a fly. Don't raise both arms). Once it has evaluated what should happen and what shouldn't, it sends a green light to the primary motor cortex.
3. The primary motor cortex receives the green light and executes the command.

Since the basal ganglia smoothed the signal, the wave appears controlled, intentional, and socially appropriate.

The ability for the basal ganglia to produce these effects relies on a balance of glutamatergic and GABAergic signaling. The default state of the basal ganglia is inhibition. This means, when we want to trigger a movement, we need to activate glutamate neurons to “break through” the inhibition that normally controls the basal

ganglia. This is where the SN comes in. The overall role of the SN is to modify the function of the basal ganglia. And it does so through the release of dopamine (DA). DA works to increase the sensitivity of basal ganglia neurons to glutamate, so that it is easier for glutamate to break through the inhibition to engage in wanted movement.

When DA is lost, it is much harder for excitatory signals from the premotor cortex to be passed through the basal ganglia and to the motor cortex. There is too much inhibition. Consider our example of waving at a friend. When DA in the basal ganglia is depleted, the wave becomes jerky, delayed, or exaggerated. It is these features that we see expressed in the motor disturbances in PD patients. Delayed movement often is seen as a slowing of movement termed **bradykinesia**. Movements may become much smaller and may be difficult to initiate at all resulting in a staggering of motor movements. Additionally, the resting rolling tremor that is seen as characteristic in most PD patients is due to the inability to turn off some unwanted motor movements.

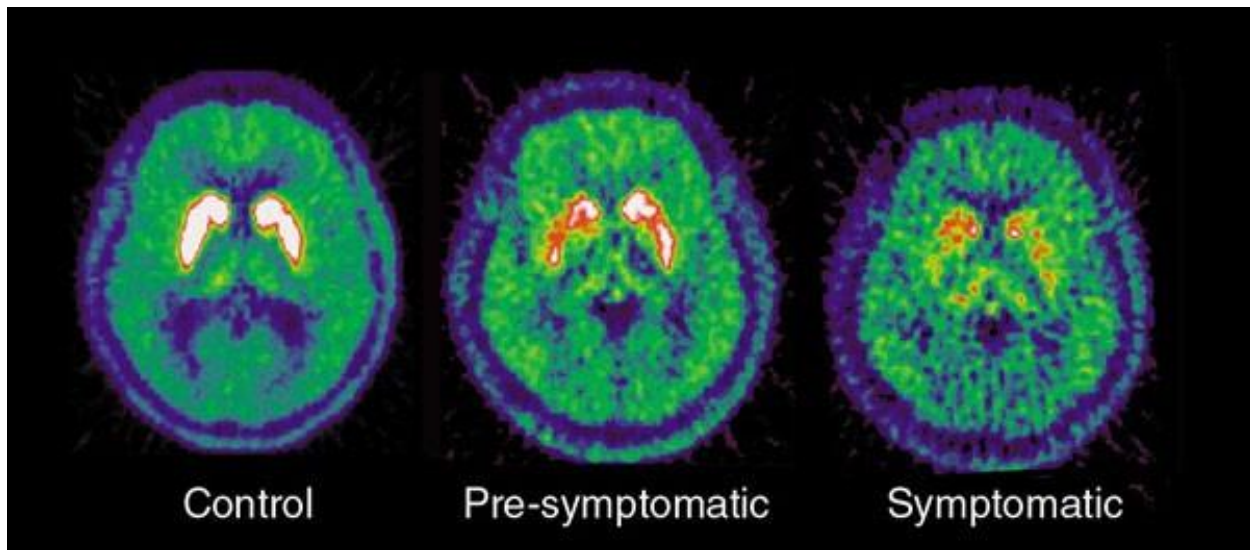
The loss of DA in PD patients is due to the progressive death of SN neurons and a slow depletion of DA.

Part II. Activity

In the next portion of this activity, we will be looking at real, published data comparing healthy patients with those who are diagnosed with PD. You will be asked questions about each figure and then asked to tie your new understanding back to your patient “James”.

[Image 1. DA binding in the basal ganglia.](#)

The image below shows PET scan data from three patients. The first is a healthy patient. The middle shows a patient who is presymptomatic but is eventually diagnosed with PD and the last is a symptomatic PD patient. PET scans use color intensity to tell us about how well a neurotransmitter is binding in the brain. White suggests strong binding and activity, whereas dark violet/blue suggest that there is little neurotransmitter activity in the region.



Data from Dunnet and Bjorklund Nature (1999)

Question 1: Consider the image above. What do you notice about DA activity in the basal ganglia. How does this differ from control → presymptomatic → symptomatic patients?

Image 2. Effects of Levodopa.

Levodopa is often used as a treatment for PD. As of right now, we don't have a good way to stop the death of SN DA neurons. For this reason, treatment options largely rely on trying to restore DA signaling by other means. The first line of treatment for PD is usually Levodopa, a precursor for DA. This treatment is chosen because Levodopa can pass into the brain from the blood stream, it can bind to DA receptors and can be converted to functional DA.

The image below shows PET scans from two patients: a healthy control (left) and a PD patient who has been taking levodopa. An intensity scale is provided for you. In this case black colorations suggest no DA activity whereas red is high DA activity.

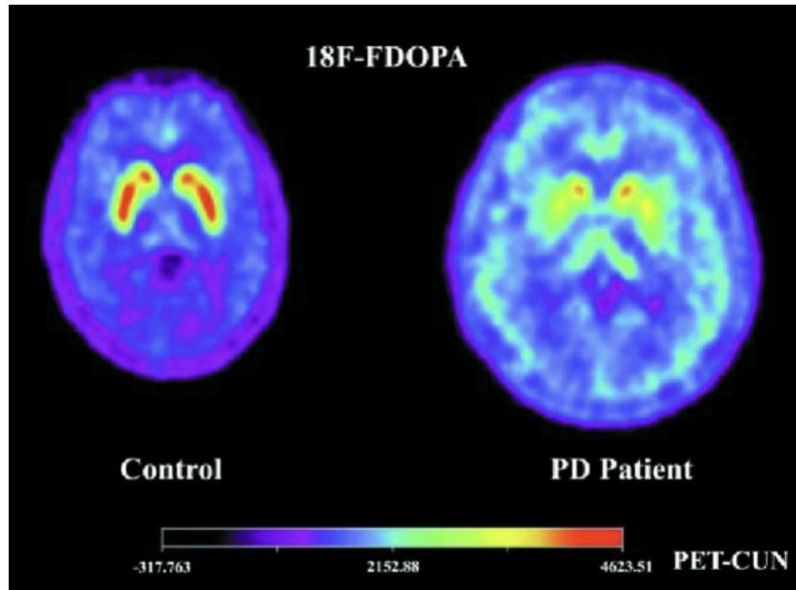


Figure taken from Obeso et al., *Physiology* 2002

Question 2: Consider the image above. What do you notice about DA activity in the PD patient? How does activity in the basal ganglia compare between healthy and control patients? Are there any notable differences?

Part III. Tying it all together

For this portion you will be asked to integrate your understanding from Parts I and II. Consider your conversation with “James” in the context of the preceding discussion and the real clinical data presented.

Question 3. Considering what you’ve learned about dopamine and the basal ganglia, explain how the loss of dopamine might be contributing the symptoms “James” is experiencing.

Question 4. Using the PET scan image from the patient on Levodopa (above), what can you infer about how effective this treatment is in restoring dopamine activity in the brain? How might this explain the side effects associated with the treatment?

Question 5. Compare the PET scans shown in Image 1: the healthy patient, the presymptomatic individual, and the symptomatic PD patient. Based on what you learned from James and these images, which stage do you think he is in? Can he be classified into a PD subtype? Explain your reasoning using both the interview and the data.

Disclosures and References.

This activity used data from real, published manuscripts on Parkinson's Disease. Full text links to each article are found below.

1. Albrecht et al., Parkinsonism & Related Disorders (2022; [full text link](#))
2. Dunnet and Bjorklund Nature (1999; [full text link](#))
3. Obeso et al., *Physiology* (2002; [full text link](#))

Note, GPT4.0 was used to select the AI bot platform and help create the persona of James Patient.